SULZER	Machine Operating Procedure (MOP) CHESTER Victory 360L Universal Milling Machine	Risk Assessment RA XXXXX

Issued By:	Prakash Giri	Region:	INMEC
Authorised By:		Sub-Region	Middle East
Applicable To:	Machine Operations	Service Centre:	SASOS Abu Dhabi

WARNING: Do not operate this machine unless you have received formal training in its safe use, have been assessed and proven competent, and are authorised to operate it through successful completion of a competency assessment.

PPE requirements

	• Safety glasses • Single piece coveralls (Short sleeves) • Safety footwear
	• Gloves must not be worn near rotating equipment • Cut-resistant gloves must be worn when handling swarf, tooling, and when loading or unloading components onto or from the lathe • Chemical-resistant gloves must be worn when changing coolant and when handling chemicals or oils

Additional safety precautions

	• When machining abnormal workpieces, a Point of Work (Dynamic) risk assessment MUST be carried out to assess the risks.		• Ensure all guarding is in place and any interlocks are working.
	• Rings and jewellery must not be worn or be covered in work areas.		• Hearing protection should be worn if the indicators are red. Ensure others in the area are complying.
	• Long and loose hair / Loose articles must be contained or secured.		• Where fitted, ensure inbuilt LEV is switched on or use mobile LEV if a lot of smoke or fume is being produced.

Significant hazards



- Entanglement with rotating parts
- Ejection of workpieces or tooling.
- Contact with cutting tools and rotating components.
- Flying swarf and metal chips.
- Manual handling of heavy components.

Lifesaving rules



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Step	Pre-operational safety checks
1	Machine-specific risk assessment available.
2	LED work light operational
3	Check emergency stop and normal controls appear serviceable; do not bypass any safety device.
4	DC injection brake functional.
5	Drawbar guard fitted, in good condition, and its interlock functional.
6	Emergency stops functional.

Step	Loading, unloading & set-up
1	Ensure the work area, table and floor are clean, dry and free from unnecessary items.
2	Ensure the machine is stopped before setup. Apply the site isolation/lockout procedure whenever hands or tools may enter a hazardous area or where unexpected start-up could cause injury
3	Clean the table, vice, fixtures and mating surfaces. Remove swarf using a brush or suitable tool, not bare hands
4	Check chuck/arbor/toolholder and cutting tool for damage, correct fit and secure retention.
5	Remove the chuck key and all setting tools before starting the spindle
6	Position the workpiece on the table/base and secure it using a vice, clamps, fixtures or suitable bolting. Never hold the workpiece by hand.
7	Confirm the intended spindle speed and feed are suitable for the material, tool diameter and operation
8	Check machine head, radial arm and column are correctly positioned and clamped before cutting
9	Arrange coolant, if required, and verify hoses will not contact rotating components
10	Do not use the machine if abnormal noise, damage, leakage, loose components or unsafe conditions are found. Tag/report the defect according to site procedure
11	Repeat until the workpiece is running within the required tolerance.
12	Remove all tools from the bed and slides of the machine.

Step	Operational safety
1	Stand in a stable position with hands clear of the cutter. Warn nearby personnel before starting
2	Start the spindle using the normal controls. Observe from a safe position for abnormal vibration, noise, loose workholding or cutter runout. Stop immediately if abnormal conditions occur
3	Faulty equipment must NOT be used. Immediately report suspect machinery.
4	Machine operator MUST NOT leave the machine unattended while operational and MUST maintain vigilance at all times.
5	Set the correct speed and feed before starting. Make speed-range or drive changes only as permitted by the machine manufacturer and with the machine stopped where required.
6	Avoid letting swarf build up on the tool or job. Stop the machine and remove it using a swarf hook or brush. Wear cut-resistant gloves.
7	Apply feed smoothly. Do not force the tool. Monitor the cut, work holding, sound, vibration and chip formation
8	Ensure all applicable guards are in place while operating the machine.
9	Use a brush, chip hook or suitable tool to remove swarf only when it is safe to do so. Never clear swarf with bare hands
10	Stop the spindle before measuring, checking the hole, repositioning/clamping the work, changing tools, adjusting the workpiece or leaving the machine unattended

Step	Housekeeping
1	Switch off the machine and reset all guards to a fully closed position.
2	Leave the machine in a safe, clean, and tidy state.